

Arkobrato Gupta

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EDUCATION

The Ohio State University

Columbus, OH, USA

- *Ph.D. in Biostatistics & M.S. in Statistics*

Aug 2021 - Aug 2027 (expected)

- Advisor: Dr. Dongjun Chung
- Dissertation: Statistical frameworks integrating multi-modal biological data, with applications in immuno-oncology and neurogenomics of opioid use disorder
- Coursework: Advanced Survival Analysis, Statistical Genetics, Causal Inference, Bayesian Inference, MCMC, Bioinformatics & Machine Learning in Omics

Hindu College, University of Delhi

New Delhi, India

- *M.S. in Statistics*

Jul 2017 - Jun 2019

St. Xavier's College, University of Calcutta

Kolkata, India

- *B.S. (Hons.) in Statistics*
- Supervisor: Durba Bhattacharya
- Thesis: Application of Bayesian hierarchical modeling in astro-statistics
- Fit a Bayesian hierarchical Poisson change-point model to Chandra X-ray flare data from young stars using hybrid Gibbs/Metropolis-Hastings MCMC in R

Jul 2014 - Jun 2017

RESEARCH EXPERIENCE

The Ohio State University Comprehensive Cancer Center

Columbus, OH, USA

Graduate Research Associate

May 2022 - Present

Pelotonia Institute for Immuno-Oncology

Dissertation research

- **SRT:** Developed a statistical framework integrating spatial TCR-seq and spatial transcriptomics (earth-mover's distance + Gaussian-mixture FDR control); identified 878 genes coupled with T-cell clonal diversity in glioblastoma, enabling biomarker discovery in the tumor microenvironment.
- **Opioid use disorder** (with multi-site collaborators): Developed Bayesian stochastic block model clustering for 900+ genetically diverse rats, classifying heroin-use-disorder vulnerable, intermediate, and resilient phenotypes and revealing sex-specific subgroups within the vulnerable phenotype (*American Journal of Psychiatry*, 2025). Extending the framework to link behavioral subgroups with brain-region gene expression.
- **SCOPE:** Built an R Shiny app implementing this clustering framework (modularity-initialized Gibbs sampling with posterior uncertainty quantification); adopted by 5+ labs in the USA, Italy, and the UK.

Collaborative projects

- **DNA methylation:** Co-led a genome-wide EWAS (Illumina EPIC; 802,978 CpGs; $n = 324$ postmenopausal women) identifying 736 differentially methylated loci between Native Hawaiian and White women, associated with liver disease pathways (61% of genes), DASH diet adherence, and adiponectin (*Scientific Reports*, 2025; co-first author).
- **Comparative oncology:** Performed statistical analyses of clinical, histopathologic, and gene expression data from 75 dogs with soft tissue sarcoma; higher FOXP3⁺ regulatory T-cell infiltration was associated with shorter progression-free intervals, and tumors resembled human undifferentiated pleomorphic sarcoma, supporting canine sarcoma as a preclinical model (*Cancers*, 2025).

The Ohio State University, Columbus, OH

Indian Institute of Management (IIM) Ahmedabad**Ahmedabad, India***Research Intern**Aug - Sep 2018*

- Collaborated with a multidisciplinary team to quantify public support for nuclear energy using survey data and regression analysis.

National Sample Survey Office (NSSO), Govt. of India**Kolkata, India***Research Intern**Jun - Jul 2018*

- Analyzed national sample survey data using logistic regression and survey methods to identify determinants of catastrophic health expenditure in India; presented findings to the Ministry.

INDUSTRY EXPERIENCE

Abbott Nutrition**Columbus, OH, USA***Biostatistics Intern**Summers 2025 & 2026*

- **2025:** Analyzed a propensity-matched NHANES cohort ($n = 6,508$; 2011–2018): adults consuming ≥ 1.2 g/kg/day protein had 1.2% greater lean soft tissue ($p < 0.001$), independent of physical activity, supporting protein guidelines for healthy aging.
- Owned the full analysis lifecycle: cohort definition, statistical method selection, confounder adjustment, and inference; communicated results to scientific stakeholders and presented to leadership. Accepted for poster presentation at ICFSR 2026 (Washington, DC).
- **2026:** Conducted an individual participant data (IPD) meta-analysis of 14 clinical trials using both two-stage and one-stage (mixed-effects) approaches, and explored mixture models to capture heterogeneity in intervention effects on key health outcomes; presented results to the team.

Novartis Healthcare Pvt. Ltd.**Hyderabad, India***Statistical Programmer**Aug 2019 - Jul 2021*

- Programmed SDTM and ADaM datasets and tables, listings, and figures (TFLs) in SAS and R to CDISC standards for three clinical trials, including the pivotal Phase III study of Kisqali (ribociclib) in breast cancer.
- Implemented the statistical methods specified in statistical analysis plans (SAPs), working with trial statisticians on analysis specifications and outputs.
- Performed independent double programming and quality-control validation of datasets and outputs to ensure accuracy and submission readiness.
- Built reporting databases supporting regulatory submissions across solid-tumor oncology and respiratory therapeutic areas.

PUBLICATIONS

Peer-Reviewed Papers

* Joint first authors

- Song, M. A.*, **Gupta, A.***, Park, S. Y., Kim, J., Archer, K. J., Tiirikainen, M., Franke, A., Wilkens, L. R., Le Marchand, L., Chung, D., & Lim, U. (2025). Differential blood DNA methylation loci between Native Hawaiian and White women are associated with dietary patterns and metabolic biomarkers. *Scientific Reports*, 15, 43191. doi:10.1038/s41598-025-25253-4
- Kuhn, B. N., Cannella, N., Crow, A. D., Lunerti, V., **Gupta, A.**, Walterhouse, S. J., ..., Chung, D., Ciccocioppo, R., & Kalivas, P. W. (2025). Distinct behavioral profiles and neuronal correlates of heroin vulnerability versus resiliency in a multi-symptomatic model of heroin use disorder in rats. *American Journal of Psychiatry*, 182(2), 198–208. doi:10.1176/appi.ajp.20230623

- Kuhn, B. N., Cannella, N., Chitre, A. S., ..., **Gupta, A.**, ..., Chung, D., ... & Palmer, A. A. (2025). Genome-wide association study reveals multiple loci for nociception and opioid consumption behaviors associated with heroin vulnerability in outbred rats. *Molecular Psychiatry*, 30(8), 3363–3375. doi:10.1038/s41380-025-02922-4
- Hayburn, R., Chung, D., **Gupta, A.**, Sills, S., Wu, J. D., Ambrus, A., Jing, W., Ng, J., Penaloza-MacMaster, P., Selmek, A., Pollack, S. M., & Bracha, S. (2025). The immune landscape of canine soft tissue sarcomas as a model for human soft tissue sarcomas. *Cancers*, 17(23), 3860. doi:10.3390/cancers17233860
- Deng, Q.^{*}, **Gupta, A.**^{*}, Jeon, H.^{*}, Nam, J. H., Yilmaz, A. S., Chang, W., Pietrzak, M., Li, L., Kim, H. J., & Chung, D. (2023). graph-GPA 2.0: improving multi-disease genetic analysis with integration of functional annotation data. *Frontiers in Genetics*, 14. doi:10.3389/fgene.2023.1079198
- Jeon, H., Xie, J., Jeon, Y., Jung, K. J., **Gupta, A.**, Chang, W., & Chung, D. (2023). Statistical power analysis for designing bulk, single-cell, and spatial transcriptomics experiments: review, tutorial, and perspectives. *Biomolecules*, 13(2), 221. doi:10.3390/biom13020221

Preprints

- Shin, J., Yoo, J., Cho, Y., **Gupta, A.**, Mao, P.-H., Thakkar, K., Eddy, T. A., Chung, K. J., Kim, J., Jeon, H., Xie, J., & Chung, D. (2026). Circular data analysis for spatial omics. *bioRxiv*. doi:10.64898/2026.09.29.755467
- Walterhouse, S. J., Crow, A. D., Dereschewitz, E., Hopkins, J. L., **Gupta, A.**, Hodebourg, R., Chung, D., Kalivas, P. W., Garcia-Keller, C., & Kuhn, B. N. (2026). Attenuation of nucleus accumbens dendritic spine density following protracted abstinence in heroin vulnerable rats. *Preprint*.
- Nagy, G., Presley, R., Dhar, S., Jeon, H., Moreira, L., Tull, O., **Gupta, A.**, Chung, D., Tschlis, P. N., & Parvin, J. D. (2024). Identification of TAP2 protein variants resistant to inhibition by the HSV1 ICP47 protein. *bioRxiv*. doi:10.1101/2024.11.18.624061
- Nagy, G., Presley, R., Jeon, H., ..., **Gupta, A.**, Banerjee, T., Chung, D., Tschlis, P. N., & Parvin, J. D. (2024). Multiplexed functional analysis of TAP2 variants in regulating MHC-I cell surface abundance reveals overexpression of PLK1 downregulates antigen presentation. *bioRxiv*. doi:10.1101/2024.09.30.615919

PRESENTATIONS

- **Gupta, A.** (2026). *Joint Statistical Meetings (JSM) 2026*. Oral Presentation: SRT, a statistical framework integrating spatial TCR-seq and transcriptomics (dissertation research).
- **Gupta, A.**, et al. (2026). *International Conference on Frailty & Sarcopenia Research (ICFSR) 2026*, Washington, DC. Poster (presented by a co-author): Protein intake and lean soft tissue in a propensity-matched NHANES cohort.
- **Gupta, A.** (2026). *Midwest Biopharmaceutical Statistics Workshop (MBSW) 2026*. Poster Presentation: SRT.
- **Gupta, A.** (2026). *ENAR Spring Meeting 2026*. Oral Presentation: SRT (dissertation research).
- **Gupta, A.** (2026). *Annual Joint Biostatistics Symposium 2026*, Columbus, OH. Poster Presentation: SRT and graph-GPA 2.0.
- **Gupta, A.** (2023). *ICSA Midwest Chapter & NIC-ASA Joint Fall Meeting 2023*. Poster Presentation: graph-GPA 2.0: improving multi-disease genetic analysis.

- **Gupta, A.** (2023). *Annual Joint Biostatistics Symposium 2023*, Columbus, OH. Poster Presentation: SRT and graph-GPA 2.0.

AWARDS & SCHOLARSHIPS

- Graduate Career Accelerator Fund (CAF) Award, The Ohio State University (May 2026)
- MBSW Student Travel Grant (May 2026)
- Koch Graduate Student Travel Award, The Ohio State University (Mar 2026)
- ICSA Student Travel Grant (Oct 2023)
- INSPIRE SHE Scholarship, Department of Science & Technology, Govt. of India (2014 - 2019)
Five-year award for students in the top 1 percentile

RESEARCH GROUP ACTIVITIES

- **Immuno-Oncology Informatics Group (IOIG), Pelotonia Institute**
Contributed to the institute's immuno-oncology research services through data analysis, statistical consultation, and training; presented project updates at monthly group meetings.
- **Chung Research Group, The Ohio State University**
Regularly presented at lab meetings, giving detailed updates on research progress, reviewing relevant literature, and leading discussions among lab members.

LEADERSHIP & SERVICE

- **Student Representative, Columbus Chapter, American Statistical Association (ASA)** (2026 - 2027)
- **President, Statistics & Biostatistics Graduate Student Association, The Ohio State University** (Aug 2024 - Aug 2025)
Led a 100+ member organization; liaised between graduate students and department administration.
- **Officer, Public Health Graduate Student Association (PHGSA), The Ohio State University** (Aug 2022 - Aug 2023)
Served as an officer of the graduate student association for the College of Public Health.
- **Placement Coordinator, Department of Statistics, University of Delhi** (Jan 2018 - Jun 2019)
Coordinated full-time placements for 70+ students and internships for 70+ students; invited, hosted, and communicated with recruiting companies, and helped train students for the recruitment process.

TEACHING EXPERIENCE

The Ohio State University

Instructor & Graduate Teaching Associate

Division of Biostatistics, College of Public Health

- Instructor, PUBHBIO 2210: Biostatistics for Public Health Research (undergraduate)
Teaching two sections (50 students in total)
- Graduate Teaching Associate, PUBHBIO 7255: Introduction to Causal Inference (graduate)

Columbus, OH, USA

Aug 2026 - Present

The Ohio State University

Graduate Teaching Associate

Division of Biostatistics, College of Public Health

- PUBHBIO 2210: Biostatistics for Public Health Research (undergraduate)
- PUBHBIO 6270: Introduction to SAS for Public Health Students (graduate)

Columbus, OH, USA

Aug 2021 - May 2022

SOFTWARE

- **SCOPE:** R Shiny app for Bayesian stochastic block model clustering with posterior uncertainty quantification; used by 5+ labs in the USA, Italy, and the UK.
- **SRT:** Statistical framework for integrating spatial TCR-seq and transcriptomics data.

SKILLS

- **Clinical & Regulatory:** CDISC (SDTM, ADaM, TFL), Phase I-III trial programming, survival analysis, causal inference, mixed models
- **Omics & Genomics:** Seurat, DESeq2, Cell Ranger (10x Genomics), PLINK (GWAS), spatial transcriptomics, TCR-seq, DNA methylation
- **Statistics & Machine Learning:** Bayesian inference, MCMC, FDR control, deep learning, feature selection, Gaussian mixture models
- **Programming & Software:** R (Shiny, R Markdown, Quarto), SAS, LaTeX, VS Code; also C, C++, Stata, SPSS, SQL

REFERENCES

Available upon request.